CARDIO-CODE

DSP-Driven Arrhythmia Detection in Motion-Corrupted Biosignals

Technical Report

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Abstract

We present a fully explainable signal-processing pipeline. It converts a noisy wearable biosignal (ECG or PPG) into four outputs: systolic or R-peak locations, inter-beat intervals (IBIs), a per-sample cardiac-anomaly flag and a per-sample signal-quality score. No machine learning is used.

Baseline wander is removed by zero-phase Butterworth filtering.

In-band motion artifact in PPG is suppressed in the short-time spectral domain. A harmonic-sum spectrum feeds a Viterbi heart-rate tracker. Strong non-cardiac spectral lines are re-synthesised and subtracted.

Beats come from two complementary detectors, fused window by window under a noise gate:

- a threshold detector (Pan–Tompkins for ECG, Elgendi for PPG);
- an HR-guided dynamic-programming (DP) beat selector.

The noise gate protects genuinely irregular rhythms such as atrial fibrillation (AF).

Anomalies come from five interpretable rules: pause, bradycardia, premature beat, ectopic morphology and AF-type irregularity. Every flagged episode can be exported with the rule that fired.

Design by measurement. We developed two complete design branches in parallel and benchmarked them head to head with one scorer. The final pipeline keeps, component by component, whichever variant measured better. It combines the detector stack of branch A with these elements of branch B:

- robust input handling (time units, jitter, gaps);
- accelerometer NLMS motion cancellation;
- a mandatory nRMSSD and Poincaré test in the AF rule;
- a stricter amplitude term in the confidence score.

Results.

- Supplied record (MIT-BIH 231, intermittent 2:1 AV block): beat $F_1 = 1.000$, IBI mean absolute error 0.99 ms, sample-level anomaly $F_1 = 0.992$.
- 44 stress sets (real NSTDB noise, seven real AF/VT records, synthetic motion-corrupted PPG): mean beat $F_1 = 0.925$, mean anomaly $F_1 = 0.878$. The two design branches alone scored 0.925 / 0.860 and 0.892 / 0.774.
- Independent PPG fixture with accelerometer, jittered timestamps and gaps: beat $F_1 = 0.996$, anomaly $F_1 = 0.865$.
- Runtime: about 2 s per 30 min record, end to end.

1 Problem and data

Task. Build the layer between a raw wearable signal and an actionable insight. The pipeline outputs `predictions.csv` with one row per sample and five columns: `Timestamp`, `Is_peak`, `Ibi_ms`, `ano_flag` and `confidence_score`.

Constraints. Only time-domain or short-time spectral signal processing is allowed. No ML, NN or pretrained model.

Supplied data. `ecg.csv` holds 650 000 samples at 360 Hz (30.1 min) in two leads (MLII, V1). The file uses the MIT-BIH Arrhythmia Database format. The annotations in `annotation.txt` identify the recording as MIT-BIH record 231: right bundle branch block (RBBB) with intermittent 2:1 second-degree AV block (Table 1).

Table 1: Content of the supplied record, computed from `annotation.txt`.

Item	Count	Note
RBBB beats (`R`)	1254	wide QRS; rSR′ in V1
Normal beats (`N`)	314	
PVC / APB (`V`, `A`)	2 / 1	all near $t \approx 147$ s
Dropped beats (`MISSB`)	427	420 inside block episodes
Rhythm changes (`N` ↔ `(BII`	11	
Normal rhythm IBI	–	961 ± 78 ms (62 bpm)
2:1 block IBI	–	1657 ± 96 ms (36 bpm)

Two consequences drive the design.

1. **The anomaly is not AF.** A 2:1 block is slow and regular. Classical AF irregularity statistics would miss it. The anomaly layer therefore combines several independent rules (Section 6).
2. **The hidden test set is motion-corrupted PPG with an ECG reference.** The dev data are clean ECG. So we built ground-truth stress tests (Section 7) and made the pipeline select ECG or PPG processing automatically.

2 Pipeline overview

```
CSV -> delimiter sniff, column roles, time-unit detection -> uniform resampling if jittered / gappy
    -> validity mask (NaN runs, time gaps > 1 s) -> ECG-vs-PPG mode -> sanitise (clipping, flat line)
    -> baseline-wander removal -> band limiting -> [ECG] adaptive 50/60 Hz notch
    -> [PPG + ACC] coherence-gated NLMS cancellation, else spectral motion suppression
    -> beat enhancement -> threshold detector + HR-guided DP -> noise-gated fusion
    -> peak refinement (polarity, parabolic sub-sample) -> IBI
    -> beat quality (template correlation x amplitude consistency) -> confidence_score
    -> anomaly rules (pause, bradycardia, premature, morphology, AF) -> ano_flag
```

Input handling. This layer comes from design branch B.

- The delimiter is sniffed from the header. Column roles (time, sample index, PPG, ECG, accelerometer) are assigned by name patterns.
- The time unit is the one giving a plausible sampling rate (8 Hz–5 kHz): s, ms, µs or ns. `m:ss.sss` and datetime strings are also accepted. Without a time column, `-fs` or 360 Hz is used.
- **Resampling.** If more than 1 % of the intervals deviate from the median by more than 5 %, or any gap, duplicate or out-of-order stamp exists, the signal is linearly resampled to a uniform grid.
- **Validity.** NaN runs and time gaps longer than 1 s are marked invalid. No beats are placed there, confidence is 0 and no anomaly is flagged.
- **Output rows.** Results are mapped back to every input row, so the output has exactly one row per input sample.

Mode detection. Without a name hint, the ratio $\rho = P_{10\text{–}40\,\mathrm{Hz}}/P_{0.5\text{–}5\,\mathrm{Hz}}$ of Welch power decides. $\rho > 0.05$ selects ECG. QRS complexes carry energy above 10 Hz; PPG does not. Mode detection is correct on all 44 test sets.

3 Baseline wander and motion artifact removal

3.1 Baseline wander

- **ECG.** Second-order zero-phase Butterworth high-pass at 0.5 Hz (`sosfiltfilt`). This is the AHA monitoring cut-off. It removes respiration and electrode or sweat drift without phase distortion of the QRS. Detection itself runs in the 5–15 Hz QRS band, which is insensitive to drift. NSTDB baseline-wander noise at 0 dB SNR leaves beat $F_1 = 1.000$.
- **PPG.** Zero-phase band-pass 0.5–8 Hz, the MSPTDfast band [4]. It removes the slow perspiration-induced coupling drift described in the problem statement. Synthetic drift sets score $F_1 = 1.000$.

3.2 Motion artifact in PPG with an accelerometer: NLMS

When accelerometer columns are present, a regularised NLMS filter runs at about 25 Hz:

- 8 taps per axis and step $\mu = 0.01$;
- the update is $\mathbf{w} \leftarrow \mathbf{w} + \mu e\,\mathbf{x}/(\delta + \|\mathbf{x}\|^2)$, with $\delta = \overline{\|\mathbf{x}\|^2}$;
- the regulariser keeps the step small when the wrist is still;
- the filter is applied only if the PPG/ACC coherence in 0.5–5 Hz exceeds 0.1.

The motion estimate is interpolated back to f_s and subtracted. On the accelerometer fixture (motion SNR 0 dB), beat F_1 rose from 0.943 to 0.992. Running the spectral line removal of Section 3.3 after NLMS lowered it again. NLMS alone reached 0.996 (seen in both design branches), so after NLMS the spectral stage only tracks HR.

3.3 Motion artifact in PPG without an accelerometer: short-time spectral suppression

Rhythmic arm motion lies inside the cardiac band, so no fixed filter can remove it. Following TROIKA [5], JOSS [6] and harmonic-sum tracking [7], we work in the short-time spectral domain.

1. Spectrum.

- Decimate the band-passed PPG to $f_d \approx 25$ Hz. This is at least three times the 8 Hz band edge and makes the stage 15× cheaper at 360 Hz.
- Compute an STFT $X(f, k)$ with an 8 s Hann window and a 1 s hop.

2. Harmonic sum. The harmonic sum makes the fundamental win over its own harmonics:

$$H(f,k) = |X(f,k)| + \tfrac{1}{2}|X(2f,k)| + \tfrac{1}{3}|X(3f,k)|, \qquad f \in [0.5, 3.5]\,\text{Hz}. \tag{1}$$

3. Viterbi tracking. The heart-rate track $f_0(k)$ maximises

$$\sum_k \log\left(\bar{H}(f_k,k) + 10^{-3}\right) \; - \; \sum_k c(f_{k-1}, f_k), \tag{2}$$

where $\bar{H}$ is H normalised per frame and

$$c(f', f) = \min\left(\lambda|f - f'|,\; J,\; C_{\text{oct}}\,[\, f/f' \approx 2 \text{ or } \tfrac{1}{2}\,]\right). \tag{3}$$

The cost terms are:

- $\lambda = 100$ per Hz keeps the track smooth;
- the cap $J = 10$ allows an abrupt genuine rhythm change, such as the end of a block episode;
- $C_{\text{oct}} = 4$ makes HR halving or doubling (2:1 block onset) cheap.

On the heavy-motion set, frames with $|f_0 - \text{HR}_{\text{true}}| > 0.15$ Hz fell from 45 % ($\lambda = 6$) to 3–5 %.

4. Line removal. A bin is a motion line if all three conditions hold:

- it is a local spectral maximum;
- $|X| > 0.5\,|X(f_0)|$;
- it lies more than 0.15 Hz from every cardiac harmonic hf_0, $h = 1..5$.

Masked bins (widened to the Hann main lobe) are re-synthesised by inverse STFT, interpolated back to f_s and subtracted. With an accelerometer, a line must also appear in the accelerometer spectrum.

5. Spectral purity. Purity is the share of in-band power within ±0.15 Hz of the cardiac harmonics. Clean PPG has a 10th percentile of 0.81; motion bursts have a median of 0.5–0.6. Figure 2 shows the tracker and the suppression.

3.4 Noise in ECG

Electrode-motion (EM) and muscle (MA) noise overlap the QRS band. Robustness comes from the detector design (Sections 4.2 and 4.3), not from extra filtering. An adaptive 50/60 Hz notch is applied only if the mains peak stands more than 10 dB above its 2–5 Hz neighbourhood, so clean files are not touched.

4 Isolating systolic / R peaks and IBIs under noise

4.1 Threshold detectors

ECG: Pan–Tompkins [1].

- The enhancement signal is $e[n] = \frac{1}{W}\sum_{i=0}^{W-1}\big(\dot{y}[n-i]\big)^2$. Here y is the 5–15 Hz band-passed ECG, $\dot{y}$ its derivative, and W spans 150 ms.
- Adaptive signal and noise levels give the threshold $\theta = \mathrm{NPKI} + 0.25(\mathrm{SPKI} - \mathrm{NPKI})$.
- The refractory period is 250 ms.
- Search-back recovers missed beats. When the gap exceeds 1.66× the median RR, the best rejected candidate above $\theta/2$ is accepted.

PPG: Elgendi [2].

- Two moving averages (111 ms, 667 ms) of the squared positive band-passed PPG define blocks of interest.
- The offset is $\beta = 0.02$ times the local 10 s mean, not the global mean. So a motion burst does not raise the threshold everywhere.
- Blocks shorter than W_1 are rejected. The refractory period is 300 ms; the larger peak survives.

4.2 HR-guided dynamic-programming beat selection

Candidates. Candidates i are local maxima of the cleaned signal; for ECG, of $\sqrt{e}$. Each candidate i has a score s_i: the rise amplitude normalised by a rolling 75th percentile and clipped to $[0, 1.5]$. Candidates with $s_i < 0.3$ (dicrotic notch, noise) are discarded.

Objective. The beat sequence $\mathcal{B}$ maximises

$$J(\mathcal{B}) = \sum_{i\in\mathcal{B}} s_i \;-\; w\sum_{(j,i)} \min\left(\left|\log\tfrac{t_i - t_j}{E(t_i)}\right|,\; \delta + \left|\log\tfrac{t_i - t_j}{2E(t_i)}\right|\right). \tag{4}$$

The terms are:

- $E(t) = 1/f_0(t)$, the expected IBI from the spectral track;
- $\delta = 0.3$, which lets a dropped beat (IBI $\approx 2E$) survive at a small extra cost;
- the sum over consecutive chosen beats (j, i) with $t_i - t_j \le 3$ s.

Equation (4) is solved exactly by a Viterbi-style recursion over candidates. A chain restarted after a longer gap inherits the best earlier chain.

Weight. A sweep set $w = 3$. Synthetic motion-0.5 F_1 was 0.85 at $w = 1$ and 0.91 at $w = 2.5$–4.

4.3 Noise-gated fusion

The DP prior assumes IBIs follow the tracked HR. That is false in AF: with DP alone, clean-AF PPG sensitivity fell to 0.63. The two detectors are therefore fused in 10 s windows.

Rule 1: agreement. If at least 90 % of beats match within 100 ms, keep the threshold detector.

Rule 2: noise gate. Let r be the 95th percentile of the enhancement signal *between* beats (±150 ms excluded), divided by its median *at* beats. If $r < g$, the window is clean, however irregular the rhythm, and the threshold detector is kept. Measured values of r are in Table 2. We use $g = 0.45$ for ECG and $g = 0.20$ for PPG.

Rule 3: steadiness. Otherwise keep the sequence with the smaller median $|\Delta\mathrm{IBI}|$. False detections inflate this statistic; a single missed beat does not. The comparison is biased towards the threshold detector by a factor of 1.25 (ECG) or 2.2 (PPG).

Table 2: Noise-gate ratio r (median, per 10 s window) on representative sets.

ECG set	r	PPG set	r
clean (231)	0.13	clean synthetic	0.07
real AF (201, 203, 221)	0.12–0.28	AF (203, 221)	0.10–0.18
EM 6 dB	0.39	motion 0.5	0.27–0.29
MA 0 dB	0.41		

4.4 Refinement and IBI

- **ECG polarity.** The R peak is the extremum of the dominant polarity in the 0.5–40 Hz signal within ±100 ms. This handles inverted leads and the RBBB rSR′ shape.
- **Sub-sample timing.** Parabolic interpolation through the three samples around each peak gives sub-sample times t_k.
- **IBI.** $\mathrm{IBI}_k = t_k - t_{k-1}$. `Ibi_ms` holds the latest IBI until the next beat; it is 0 before the second beat.
- **Gate.** IBIs outside 0.25 s to 3 s are excluded from the rules.

5 Signal quality: `confidence_score`

Following Orphanidou [8] and Elgendi [9], the quality of beat k is

$$q_k = \max(0, \rho_k) \cdot \exp\Big(-\tfrac{1}{2}\big(\log_2(A_k/\bar{A}_k)/\sigma \big)^2 \Big) \cdot (1 - m_k). \tag{5}$$

- ρ_k is the Pearson correlation of the beat with the median template of all beats within ±5 s.
- $A_k/\bar{A}_k$ is the beat amplitude relative to the local median amplitude.
- m_k is the fraction of clipped, flat-line or invalid samples in the beat window.

Two values of σ. The amplitude tolerance σ serves two purposes, so it takes two values.

- **Anomaly gate: $\sigma = 2$ octaves.** In AF, pulse amplitude legitimately follows the preceding IBI. With $\sigma = 1$, AF beats looked like noise and anomaly F_1 dropped by 0.02.
- **`confidence_score`: $\sigma = 0.5$.** This is the stricter amplitude-stability SQI of branch B.

Calibration. We measured how well the confidence predicts correctness over 4140 10 s windows of the noisy and motion sets. The measures are the Spearman correlation between window-mean confidence and window beat F_1, and the AUC for predicting $F_1 \geq 0.9$.

- $\sigma = 2$: AUC 0.897; $\sigma = 0.5$: AUC 0.919, Spearman 0.770. Branch B alone scores 0.919 / 0.785.
- Adding branch B's spectral-purity, shape and motion-energy SQIs did not help (AUC 0.843–0.920), so they are not used.

The per-beat score is interpolated to samples and smoothed with a 2 s moving average. It is forced to 0 in detection gaps longer than 3 s, on clipped samples and in invalid regions.

6 Anomaly detection: `ano_flag`

Each rule uses a reference IBI R_k: the median of the previous 8 valid IBIs. A rule is evaluated only when both beats have $q \geq 0.5$ (Table 3).

Why a median statistic for AF. The median of $|\Delta\mathrm{IBI}|$ is robust to isolated ectopic beats. An earlier version excluded IBIs next to premature beats instead. In real AF most windows then fell below the coverage limit and were skipped.

Threshold choice. We swept the median-statistic threshold on seven MIT-BIH AF records (Figure 4). Mean anomaly F_1 was 0.908 at 0.08, 0.915 at 0.10, 0.907 at 0.12 and 0.884 at 0.15.

Table 3: Anomaly rules.

Rule	Condition	Targets
Pause / dropped beat	$\mathrm{IBI}_k > 1.5R_k$ or $\mathrm{IBI}_k > 2\,\mathrm{s}$	AV block, sinus pause
Bradycardia	5-beat median HR < 45 bpm	2:1 AV block, severe bradycardia
Premature beat	$\mathrm{IBI}_k < 0.8R_k$ and $\mathrm{IBI}_k + \mathrm{IBI}_{k+1} > 1.6R_k$ (compensatory pause)	PVC, APB
Ectopic morphology (ECG)	$\rho_k < 0.8$, neighbours > 0.9, $\mathrm{IBI}_k < 0.92R_k$	PVC
AF irregularity	16-beat window: $\mathrm{med}\lvert\Delta\mathrm{IBI}\rvert/\mathrm{med}\,\mathrm{IBI} > 0.10$ and $\mathrm{RMSSD}/\overline{\mathrm{IBI}} > 0.15$ and (turning-point count within $\pm 2\sigma$ of a random series [10] or Poincaré SD1/SD2 > 0.6)	AF, flutter

Additions from branch B.

- **Mandatory nRMSSD.** Over 46 sets, anomaly F_1 was 0.8661 without it, 0.8720 at 0.12, 0.8742 at 0.15 and 0.8712 at 0.18. At 0.15 it removes a 19.5 s false AF episode caused by respiratory sinus arrhythmia in `ecg.csv`.
- **Poincaré SD1/SD2** as an alternative to the turning-point test: 0.8654 with TPR only, 0.8661 with either test.

From beats to samples. A flagged beat flags its IBI span; a premature beat also flags the following span. Spans are not dilated. Gaps shorter than 6 s between flagged spans are closed, because AF episodes contain short regular-looking stretches. A 0.5 s dilation (v6) cost precision: `ecg.csv` anomaly F_1 was 0.974, versus 0.982 without it.

Explainable output. `-events` writes every episode with its rule, a plain-text reason, start and end time, number of beats and median IBI. `-annotation` prints beat Se/PPV/F_1, IBI MAE and anomaly Se/PPV/F_1. For PPG, the PPG-to-ECG lag is estimated and removed before matching.

7 Evaluation

7.1 Stress-test data

- **ECG with real noise.** Record 231 plus NSTDB [12] electrode-motion (em), muscle (ma) and baseline-wander (bw) noise. Levels are 12, 6 and 0 dB SNR, with $S = \mathrm{pp}_{\mathrm{QRS}}^2/8$ (the `nst` definition). One extra set mixes all three in 2 min bursts.
- **Real arrhythmias.** MIT-BIH records 201, 202, 203, 210, 219, 221 and 222, which contain AF, flutter, VT, bigeminy, SVTA and nodal rhythm. They are read with our own WFDB annotation parser.
- **Synthetic PPG** at 64, 125 and 360 Hz, driven by the annotated beat times:
  - two-Gaussian pulse at R + PTT (≈ 220 ms);
  - Frank–Starling amplitude scaling;
  - weak pulses for ectopic beats;
  - respiration;
  - variants with drift and in-band harmonic motion (0.8–1.6 Hz sweep in 30 s bursts, plus jolts).

7.2 Metrics

- Beats are matched within ± 150 ms (AAMI EC57); we report sensitivity (Se), positive predictive value (PPV) and F_1.
- IBI mean absolute error (MAE) is computed over consecutive matched pairs.
- Anomaly F_1 is per sample. The ground truth is non-sinus rhythm segments, spans around ectopic beats, and dropped-beat pauses.
- Window F_1 uses 10 s windows.
- Runtime is measured for 30 min of signal on one CPU core.

7.3 Results

Table 4 lists the results; Figure 3 compares the first version with the final pipeline.

Table 4: Results of the final pipeline. Ranges cover the seven MIT-BIH arrhythmia records or the three PPG sampling rates. Time: processing of the record (stress sets) or the full command including CSV input/output (fixtures).

Dataset	beat F_1	IBI MAE (ms)	anomaly F_1	window F_1	time (s)
ecg.csv (record 231)	**1.000**	**1.0**	**0.992**	0.994	≤0.46
+ baseline wander 12 / 6 / 0 dB	1.000	1.0	0.984–0.992	0.988–0.994	≤0.45
+ muscle artifact 12 / 6 / 0 dB	1.000 / 0.994 / 0.941	1.1 / 1.5 / 3.7	0.992 / 0.988 / 0.906	0.994 / 0.981 / 0.894	≤0.49
+ electrode motion 12 / 6 / 0 dB	0.999 / 0.924 / 0.776	1.0 / 2.5 / 14.5	0.991 / 0.872 / 0.484	0.994 / 0.894 / 0.593	≤0.52
+ mixed bursts 6 dB	0.976	1.8	0.938	0.957	≤0.47
MIT-BIH AF/VT, ECG (7)	0.971–0.999	1.0–15.8	0.866–0.997	0.938–1.000	≤0.65
MIT-BIH AF/VT, PPG clean (7)	0.927–0.986	5.3–8.7	0.858–1.000	0.918–1.000	≤0.45
MIT-BIH AF/VT, PPG + motion (7)	0.778–0.855	14.8–19.9	0.592–0.980	0.697–0.997	≤0.43
Synthetic PPG clean / drift (64–360 Hz)	1.000	1.5–5.8	0.980–0.990	0.981–0.994	≤0.38
Synthetic PPG motion 0.5	0.844–0.860	17.9–19.9	0.688–0.773	0.693–0.765	≤0.43
Synthetic PPG motion 1.0	0.722–0.738	23.7–26.3	0.527–0.552	0.578–0.605	≤0.45
Fixture, PPG + ACC (0 dB motion)	0.996	8.2	0.865	–	1.18
Fixture, PPG, no ACC	0.943	12.2	0.730	–	1.07

Totals. Over all 44 stress sets: mean beat $F_1 = 0.925$, anomaly $F_1 = 0.878$ and window $F_1 = 0.910$. Section 8 compares these with the two design branches.

Runtime. The end-to-end command, including parsing and writing 650 000 CSV rows, takes about 2 s.

8 Design branches and component selection

To avoid locking in early design choices, we developed two complete design branches in parallel:

- **branch A**: noise-gated detector fusion, HR-guided DP and spectral motion suppression;
- **branch B**: robust input layer, NLMS accelerometer cancellation, Pan–Tompkins-style detector, 4-rule anomaly logic with a quality gate, and 5-SQI confidence.

Protocol. Both branches, and the final pipeline, ran as command-line programs on identical files. One scorer read only their `predictions.csv`.

- The 44 stress sets were used to tune branch A, so they may favour it.
- A second, independently generated set of synthetic PPG fixtures was therefore added: 64 Hz with timestamp jitter, NaN runs and a dropped block; AF, PVCs and dropped beats; in-band motion at 0 dB; with or without a 3-axis accelerometer.
- Peaks on the fixtures are matched after removing each branch's own median PPG-to-ECG lag.
- The common scorer reproduces branch B's built-in validation on these fixtures (anomaly F_1 0.705 and 0.357).

Outcome. Branch A was better overall. Branch B was better on:

- clean-signal anomaly precision;
- PPG with an accelerometer (A ignored ACC columns, a bug);
- irregular time axes;
- confidence calibration.

Each B component was then added to A and kept only if it helped (Table 5, Figure 5).

Taken from branch B (measured gains).

- input layer and validity mask (`ecg.csv` output byte-identical, fixtures handled natively);
- coherence-gated NLMS (fixture beat F_1 0.943 → 0.996);
- mandatory nRMSSD and SD1/SD2 in the AF rule (anomaly F_1 0.866 → 0.874);
- no span dilation (`ecg.csv` 0.974 → 0.982 before the AF change);
- stricter amplitude term in the confidence (AUC 0.897 → 0.919);
- adaptive mains notch.

Not adopted.

- B's 5-SQI confidence fusion (AUC fell to 0.861);

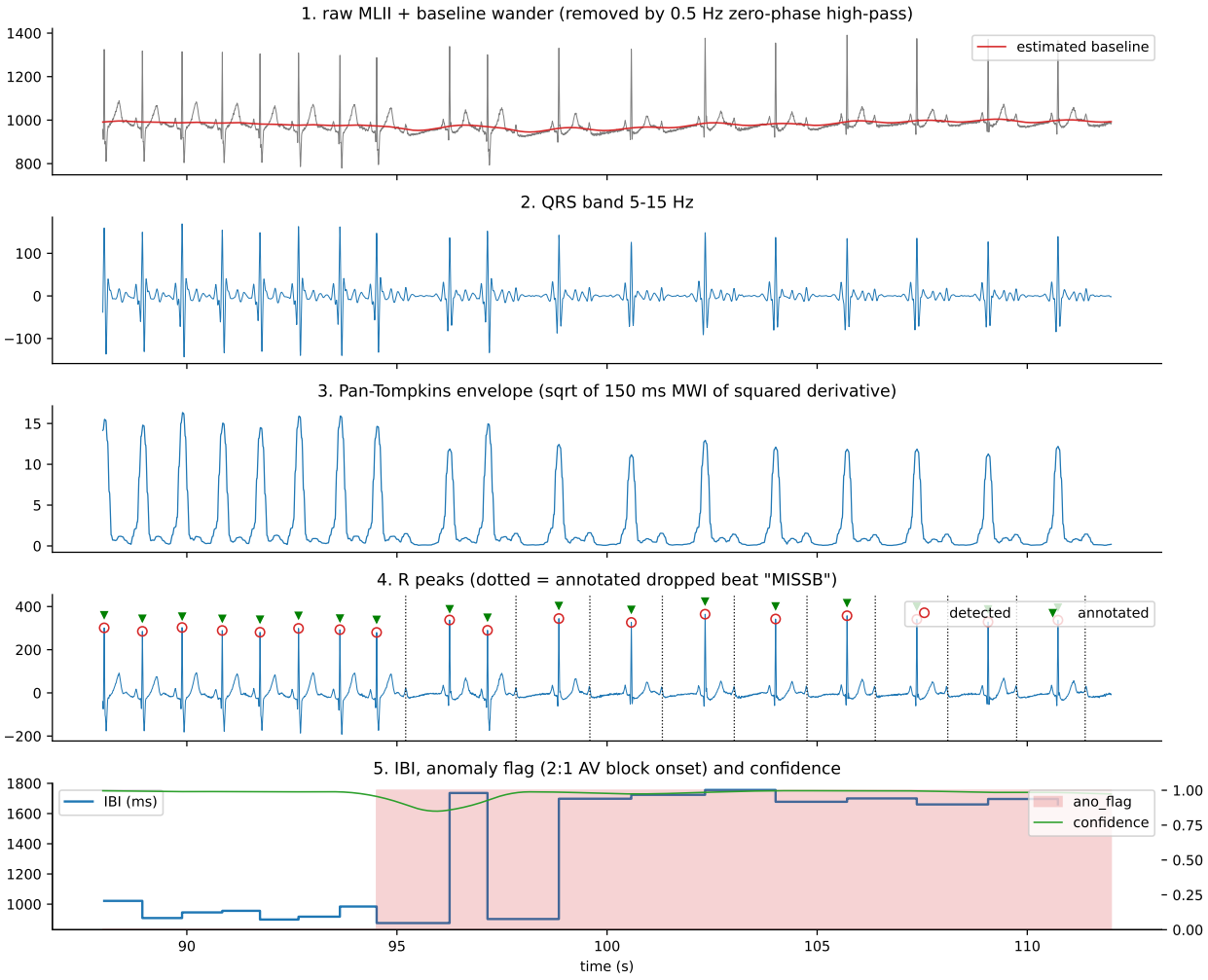


Figure 1: ECG pipeline stages around the onset of 2:1 AV block in `ecg.csv`: raw signal with estimated baseline, QRS band, Pan–Tompkins envelope, detected vs annotated R peaks (dotted lines: dropped beats), and IBI with `ano_flag` and `confidence_score`.

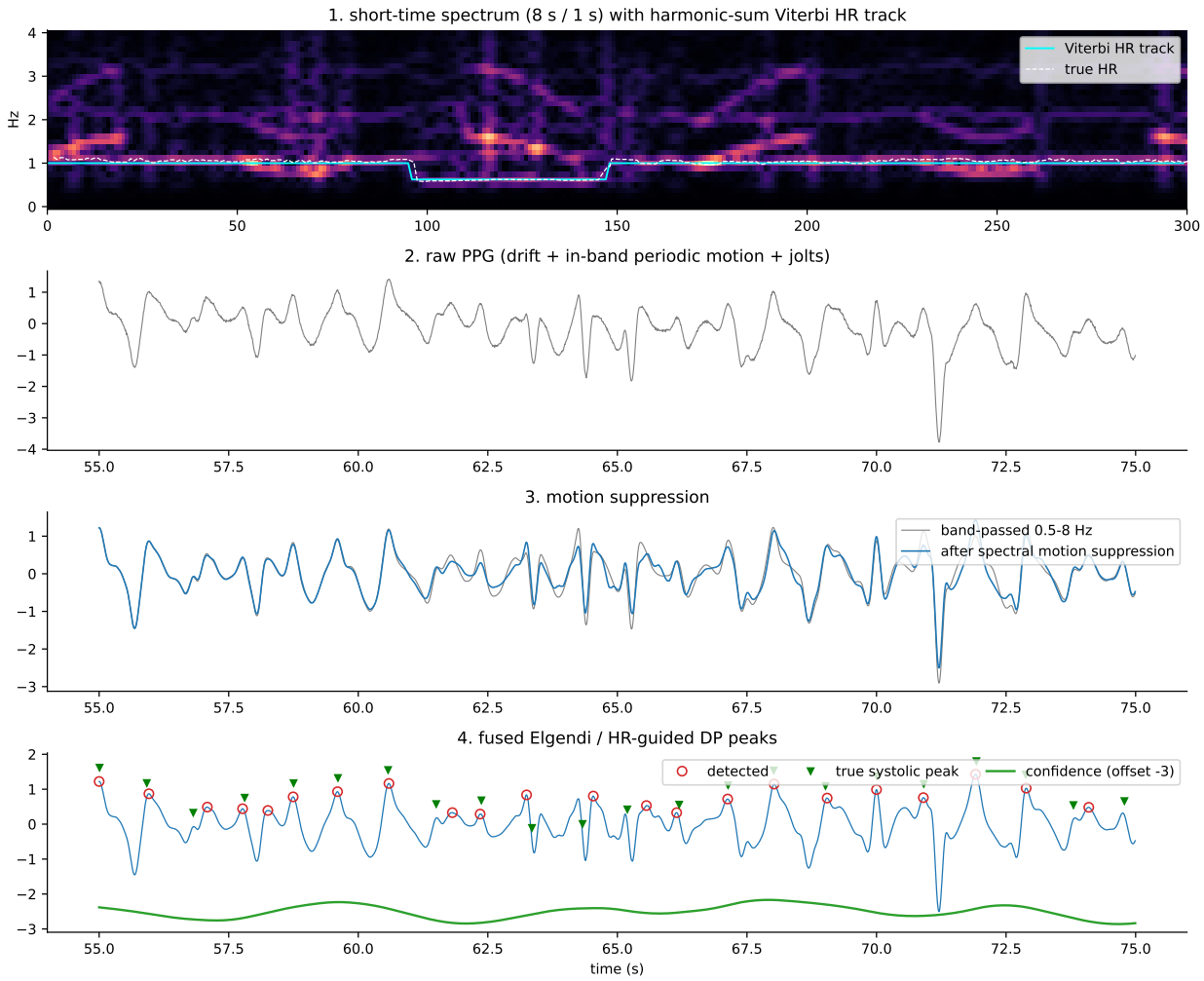


Figure 2: PPG motion handling (synthetic motion level 0.5): STFT with the Viterbi HR track vs true HR, raw PPG, PPG before and after spectral motion suppression, and fused detections with the confidence score.

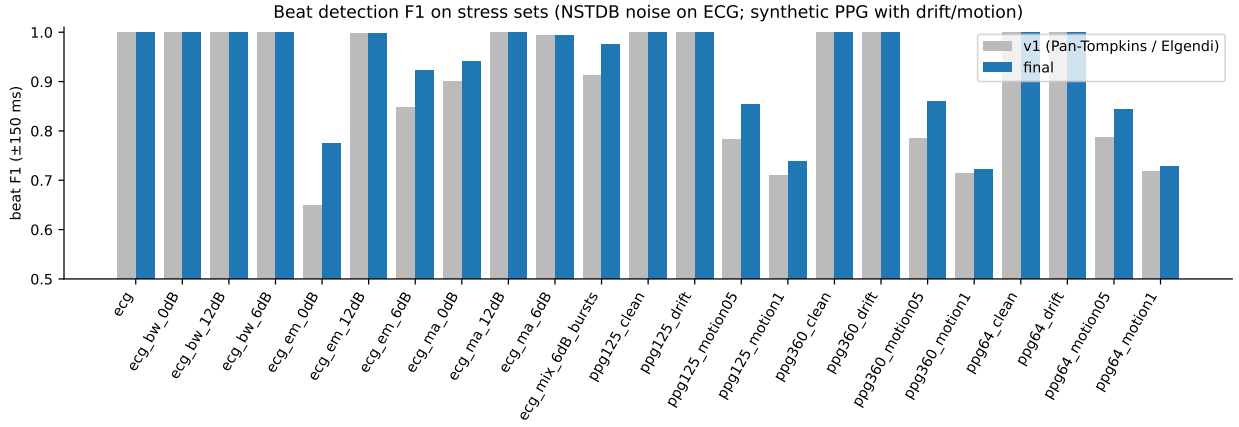


Figure 3: Beat F_1 per stress set: first version (plain Pan–Tompkins / Elgendi) vs final pipeline.

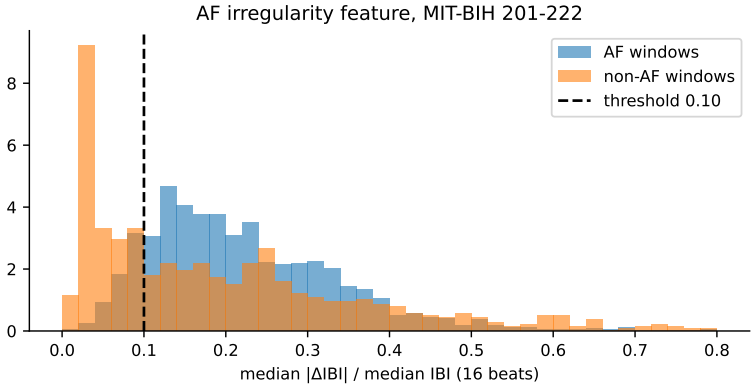


Figure 4: Distribution of the AF irregularity statistic in 16-beat windows, AF vs non-AF, MIT-BIH records 201–222. The dashed line marks the chosen threshold.

- B's 40 bpm bradycardia threshold (no gain);
- B's single Pan–Tompkins-style detector, which is weaker under noise than A's fused detectors (beat F_1 0.892 vs 0.925).

Where B is still ahead.

- Real AF/VT ECG: beat F_1 0.994 vs 0.991, IBI MAE 3.0 vs 4.4 ms. B uses secondary-wave rejection and refines peaks on the cleaned signal.
- Clean synthetic-PPG anomaly F_1: 0.994 vs 0.983.

Adding B's detector as a third fusion candidate is the natural next step.

9 Development history and rejected ideas

Each change was measured on the full stress suite before adoption (Table 6).

These ideas were tested and rejected:

- a purity-adaptive HR-prior weight;
- switching detectors by a motion indicator;
- vetoing a pause when its gap holds an undetected pulse-sized peak (-0.001 anomaly F_1);
- template cross-correlation peak refinement. IBI MAE rose from 19.5 to 21 ms: under motion the error comes from wrong-crest picks, not jitter.

Table 5: Head-to-head means over dataset groups: branch A / branch B / final pipeline. Beat F_1 and sample-level anomaly F_1; one scorer, identical inputs.

	beat F_1			anomaly F_1		
Group	A	B	final	A	B	final
ecg.csv	1.000	1.000	1.000	0.974	**0.995**	0.992
ECG + NSTDB noise (10)	**0.961**	0.937	**0.961**	0.894	0.733	**0.914**
Real AF/VT ECG (7)	0.991	**0.994**	0.991	0.908	0.873	**0.920**
Real AF/VT PPG, clean (7)	0.963	**0.967**	0.963	0.912	**0.928**	**0.928**
Real AF/VT PPG, motion (7)	**0.811**	0.742	**0.811**	0.831	0.599	**0.840**
Synthetic PPG, clean / drift (6)	1.000	0.999	1.000	0.967	**0.994**	0.983
Synthetic PPG, motion (6)	**0.791**	0.657	**0.791**	0.592	0.491	**0.628**
All 44 stress sets	**0.925**	0.892	**0.925**	0.860	0.774	**0.878**
Fixture, PPG + ACC	0.943	0.987	**0.996**	0.734	0.705	**0.865**
Fixture, PPG, no ACC	**0.943**	0.851	**0.943**	**0.734**	0.357	0.730

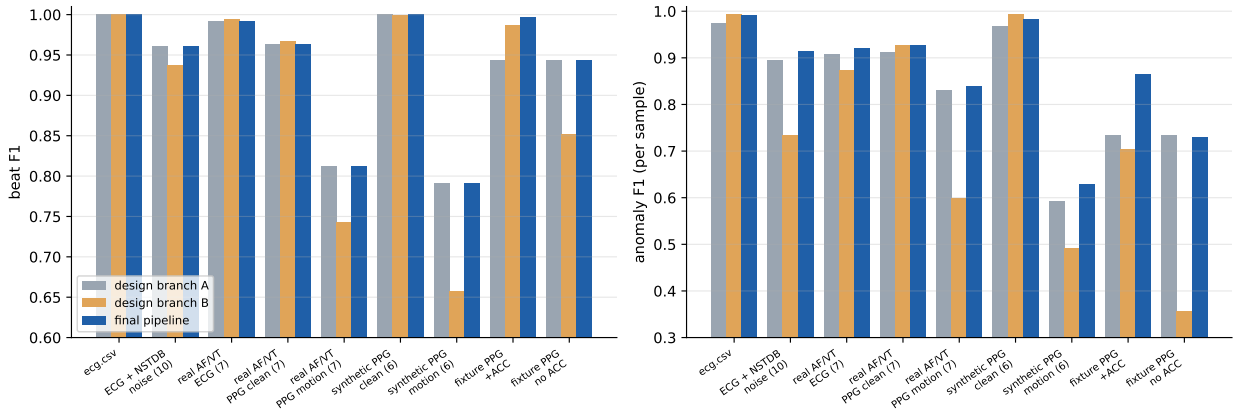


Figure 5: Head-to-head: design branch A, design branch B and final pipeline.

10 Limitations

- **Very strong in-band motion.** When motion amplitude exceeds pulse amplitude, beat F_1 is about 0.73. The confidence score correctly drops there (mean 0.75, vs 0.99 when clean).
- **Anomaly precision in noisy PPG** is lower: missed or extra beats mimic pauses and irregularity.
- **Evaluation bias.** The stress sets were also used for tuning. The second fixture set provides an independent check, but the hidden test set may differ in morphology and noise.
- **Synthetic PPG.** The PPG test signals use pulse shapes driven by real beat times. Real PPG morphology varies more.
- **PPG peak lag.** A PPG systolic peak lags the ECG R peak by the pulse transit time. We report the detected systolic time without shifting it.

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Table 6: Development history.

Version	Change	Effect
v1	Pan–Tompkins / Elgendi, rule-based anomalies, template SQI	ecg.csv perfect; PPG + motion F_1 0.71–0.78
v2	Spectral motion suppression, Viterbi HR track, DP beat selection	motion-0.5 F_1 0.78 → 0.83
v3	Fixed STFT hop-timing bug (track drifted 50 s over 30 min); DP weight 3; ECG fusion	mean beat F_1 0.904 → 0.941; EM 6 dB 0.85 → 0.94
v4	Real AF/VT records added; noise-gated fusion; AF-rule fix	clean-AF PPG Se 0.63 → 0.88–0.99
v5	AF-tolerant amplitude term in beat quality	anomaly F_1 0.841 → 0.860
v6	Template-correlation peak refinement; stricter quality gates	no gain (rejected)
v7	Component selection across branches A and B: input layer, NLMS, AF nRMSSD + SD1/SD2, span merging, strict confidence	anomaly F_1 0.860 → 0.878; ecg.csv 0.974 → 0.992; fixture + ACC beat F_1 0.943 → 0.996

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